

2024 SCHEME

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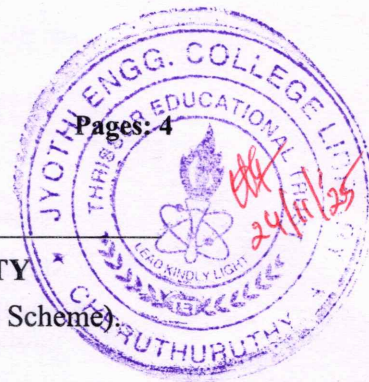
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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (R) (FT/WP) Examination November 2025 (2024 Scheme)

**Course Code: PBCST304****Course Name: OBJECT ORIENTED PROGRAMMING**

Max. Marks: 40

Duration: 2 hours 30 minutes

PART A*(Answer all questions. Each question carries 2 marks)*

- | | | CO | Marks |
|---|---|-----|-------|
| 1 | What is the role of the Just-In-Time (JIT) compiler in the Java Virtual Machine (JVM)? Explain how it improves the performance of Java programs during execution. | CO1 | (2) |
| 2 | What is the output of this code snippet and the value for n after the execution of this code?
<pre>int n = 29; // binary 11101 System.out.println(n >> 1); System.out.println(n >>> 1); System.out.println(~n);</pre> | CO1 | (2) |
| 3 | What is the output of the program, justify your answer.
<pre>// Base class (Parent) class Animal { Animal() { System.out.println("Animal constructor called"); } void sound() { System.out.println("Animals make sounds"); } } // Derived class (Child) class Dog extends Animal { Dog() { super(); System.out.println("Dog constructor called"); } void sound() { System.out.println("Dog barks"); } }</pre> | CO2 | (2) |


```
// Main class
public class InheritanceDemo {
    public static void main(String[] args) {
        Dog d = new Dog(); // Object of subclass
        d.sound();          // Calls overridden method
    }
}
```

- | | | |
|---|---|---------|
| 4 | Explain the use of the final keyword in Java. | CO2 (2) |
| 5 | What is a package in Java? Explain its advantages and types with suitable examples. | CO3 (2) |
| 6 | Differentiate interfaces and abstract classes with examples | CO4 (2) |
| 7 | How would you design a user registration GUI application that saves data to a database? Provide the Model-View-Controller (MVC) diagram representing this design. | CO5 (2) |
| 8 | Compare Java AWT and Swing. | CO5 (2) |

PART B

(Answer any one full question from each module, each question carries 6 marks)

Module -1

- | | | |
|----|--|---------|
| 9 | <p>a) Write a Java program to create an array of Student objects. Each student has the following attributes:</p> <p style="margin-left: 40px;">Name (String)
 RegNo (String)
 Marks of 3 subjects(double)</p> <p>The class should include:</p> <ol style="list-style-type: none"> 1. A parameterized constructor to initialize all attributes. 2. A default constructor that assigns default values. 3. A method <code>display()</code> to print the details of each student. <p>In the <code>main()</code> method,</p> <p style="margin-left: 40px;">Create an array of 3 Student objects.
 Use both the default and parameterized constructors to initialize the objects.
 Display the details of all students.</p> | CO2 (4) |
| | b) Explain encapsulation and polymorphism with examples. | CO1 2 |
| 10 | a) Explain Arithmetic and Bitwise operators of Java with sufficient examples | CO1 4 |
| | b) Explain the access specifiers of Java. How access specifiers help in achieving data hiding in Java? | CO2 2 |

Module -2

- | | | |
|----|---|-------|
| 11 | Write a Java program to demonstrate inheritance with the use of constructors.
The program should have the following class structure: | CO2 6 |
|----|---|-------|

Class 1: Bank (Superclass)**Attributes:**

- bankName (String)
- branch (String)

Constructor:

- Parameterized constructor to initialize the above attributes.

Method:

- displayBankDetails() — displays bank name and branch.

Class 2: SBAccount (Subclass of Bank)**Attributes:**

- accountHolder (String)
- balance (double)

Constructor:

- Parameterized constructor to initialize all attributes (including superclass attributes using super()).

Method:

- displaySBDetails() — displays complete Savings Account details.

Class 3: CurrentAccount (Subclass of Bank)**Attributes:**

- accountHolder (String)
- balance (double)

Constructor:

- Parameterized constructor to initialize all attributes (including superclass attributes using super()).

Method:

- displayCurrentDetails() — displays complete Current Account details.

Main Class: BankDemo**In the main() method:**

- Create objects for both SBAccount and CurrentAccount.
- Display their details.

- | | | | | |
|----|----|---|-----|---|
| 12 | a) | Write a Java program to demonstrate Runtime Polymorphism using method overriding. | CO2 | 4 |
| | b) | What is the use of keyword <i>Super</i> . Give example. | CO2 | 2 |

Module -3

- | | | | |
|----|---|-----|---|
| 13 | Write a Java program to create a package named College that contains a Student class with the following specifications:
The Student class should have: | CO3 | 6 |
|----|---|-----|---|
- Attributes: name (String), rollNumber (String), grade (char)
 - A parameterized constructor to initialize all attributes
 - A method displayInfo() that displays all student details
 - A method calculateGradePoint() that returns grade points (A=10, B=8, C=6, D=4, F=0)

Create a `MyClass` class in the default package that:

- 1.Imports the `College` package
- 2.Creates 2 Student objects
- 3.Displays their information and grade points

Show the complete directory structure, compilation, and execution steps.

- | | | | | |
|----|----|---|-----|---|
| 14 | a) | Differentiate between checked and unchecked exceptions in Java. Give suitable examples for each type. | CO3 | 2 |
| | b) | What is exception handling in Java? Explain the following keywords with suitable examples:
(i)try and catch
(ii)throw
(iii)throws
(iv)finally | CO4 | 4 |

Module -4

- | | | | | |
|----|----|---|-----|---|
| 15 | a) | Write a Java program using Swing components to create a simple “Login” window . | CO5 | 4 |
|----|----|---|-----|---|

The application should have the following:

- Two labels: **User ID** and **Password**
- Two text fields to input User ID and Password
- One button labeled **Submit**
- One label named **Status** (initially blank)

When the user clicks the **Submit** button, the status label should display **“Welcome”**.

- | | | | | |
|----|----|--|-----|---|
| | b) | Discuss any two java swing components. | CO5 | 2 |
| 16 | a) | What are the steps to connect to a database using JDBC? Explain with an example program. | CO5 | 4 |
| | b) | How does the Delegation Event Model work in Swing applications? | CO5 | 2 |

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